

Wildlife Live Poaching News System

Complete Technical and Operational Guide (Simple + Detailed)

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This document explains the full project in simple words, including architecture, AI, pipelines, dashboards, exports, deployment, security, and officer-level usage.

1. What this project does

This system continuously searches the internet for wildlife-crime and poaching related reports, filters noisy content, merges duplicates, enriches incidents with intelligence, and shows live insights on a command dashboard.

It is designed for operational use by wildlife officials, forest officers, crime units, and control-room teams who need a real-time situational picture instead of manually scanning hundreds of news articles.

1.1 Main output of the system

- A cleaned incident list (deduplicated): each incident is one intelligence record even if many news sites report it.
- Live alert stream for high-risk and critical incidents.
- Map + analytics for hotspots, species trends, and source quality.
- Analyst exports (CSV, PDF, Excel) and a two-sheet Excel download for Total Incidents and Source Reports Today.
- Audit-ready security and admin controls for authorized usage only.

2. Why this is useful for wildlife officials and officers

2.1 Operational value in the field

- Early warning: officers can see dangerous incidents quickly instead of waiting for manual intelligence notes.
- Coverage at scale: many providers and languages are scanned continuously, including OSINT-type feeds.
- Less noise: AI + rules remove many irrelevant items and focus on probable wildlife crime reports.
- Action orientation: each incident includes risk level, likely route hypothesis, and enforcement recommendation.
- Team coordination: map and district trends help prioritize patrols, checkpoints, and inter-agency communication.

2.2 Administrative and supervisory value

- Command view for supervisors with KPI, trend, and source confidence context.
- Daily/shift reporting through ready exports (CSV/PDF/Excel).
- Security controls and audit logs for accountable operations.
- Useful for briefings to higher officers and inter-state coordination meetings.

3. Technology stack (what is used)

- **Backend:** FastAPI (Python), SQLAlchemy ORM, APScheduler for continuous sync jobs.
- **Frontend:** React (Vite), charts with Chart.js, maps with Leaflet.
- **Database:** SQLite by default (with PostgreSQL-compatible structure patterns).
- **AI/NLP:** Hybrid engine: optional transformer zero-shot model + strong rule fallback.
- **Reporting:** openpyxl (Excel), ReportLab (PDF), CSV writer.
- **Deployment:** Railway (backend) + Vercel (frontend).
- **Security:** admin login/session token, protected APIs, rate limits, audit logs.

Important: this system can run even when heavy model dependencies are unavailable, because rule intelligence fallback remains active. This keeps operations resilient.

4. High-level architecture

Architecture follows a practical pipeline: Ingestion -> Intelligence -> Dedup/Merge -> Persist -> Alerts/Reports -> Dashboard APIs -> Frontend UI.

- Provider adapters collect raw articles from enabled sources.
- Hybrid Intelligence Engine estimates poaching relevance and extracts structured details.
- Dedup engine collapses repeat coverage into one incident row with increased report_count.
- Collector updates entities/stats/reports and raises alerts when risk triggers are met.
- APIs expose dashboard, analytics, filters, alerts, exports, OSINT, and admin operations.
- Frontend polls APIs and renders live command-center views.

5. Ingestion and search coverage

5.1 Providers and multilingual reach

The platform uses multiple providers (Google RSS, Bing RSS, GDELT, and optional key-based APIs like NewsAPI/GNews/Mediastack/NewsData), plus OSINT connectors. It supports multiple Indian languages by configurable query sets.

5.2 Aggressive coverage mode (current defaults)

- SYNC_INTERVAL_MINUTES=3
- MAX_ARTICLES_PER_QUERY=60
- MAX_QUERIES_PER_LANGUAGE=8
- PROVIDER_PARALLEL_WORKERS=8
- PROVIDER_MIN_REQUEST_INTERVAL_SECONDS=1.0
- PROVIDER_RATE_LIMIT_COOLDOWN_SECONDS=600
- AI_THRESHOLD=0.50, INDIA_THRESHOLD=0.45

These settings increase recall and web coverage. AI and rules still filter, but the system intentionally accepts broader candidate sets to avoid missing incidents.

5.3 Rate-limit resilience

- Per-provider request spacing to reduce API bans.
- 429 handling with cooldown and Retry-After logic.
- Query rotation to spread load across terms and languages over cycles.

6. Intelligence engine in simple words

The intelligence engine turns raw text into structured operational intelligence. It returns: `is_poaching`, `is_india`, `confidence`, `risk_score`, `crime_type`, `species`, `location`, `involved_persons`, `recommendation`, and `confidence explanation`.

6.1 Confidence score vs Risk score

- **Confidence score:** How sure the system is that this is a genuine wildlife-crime incident signal.
- **Risk score:** How severe/urgent this incident appears for operations if it is genuine.

So confidence answers "is this real and relevant?" while risk answers "how dangerous/urgent is it?".

6.2 How confidence is computed now

- Model poaching probability + rule score + evidence strength are blended.
- Evidence strength uses keyword intensity, species/person mentions, location quality, seizure/arrest indicators, and cross-border cues.
- Penalties are applied for false-positive patterns and strong "not wildlife crime" signal.

6.3 How risk is computed now

- Uses confidence + poaching probability as base.
- Adds severity boosts by crime type (for example horn/skin/ivory trafficking types), priority species, network and repeat indicators.
- Adds operational boosts for seizure, cross-border patterns, weapon references, quantity evidence, and multiple named persons.

6.4 Detail extraction improvements

- Location extraction improved using district/state phrase patterns (like "district of X", "Y state").
- Involved persons extraction improved with multi-name patterns and better cleanup (alias/age/noise removal).
- Operational detail extraction captures agencies, quantity mentions, FIR/case refs, vehicle refs, and money/value mentions.
- These details feed intel points and confidence explanation for analyst transparency.

7. Deduplication and incident merging

Many different articles can describe one crime. The dedupe engine avoids clutter by merging them into one incident row.

- Hard checks: URL hash/canonical URL matching.
- Soft checks: title similarity and semantic similarity.

- When duplicate found: incident is updated, report_count increases, sources are merged, strongest intelligence is retained.
- Result: cleaner dashboard and better officer decisions.

8. Data model and core tables

8.1 Core incident table: news_items

- Identity/context: title, summary, source, url, language, published_at.
- AI fields: ai_score, confidence, risk_score, ai_reason, confidence_explanation.
- Case fields: crime_type, species, state, district, location, involved_persons.
- Intelligence fields: intel_summary, intel_points, likely_smuggling_route, enforcement_recommendation.
- Dedup fields: url_hash, duplicate_confidence, source_count, report_count, merged_sources.

8.2 Supporting operational tables

- alerts: high/critical alert entries and delivery flags.
- reports: analyst-ready summarized records per incident.
- entities: extracted structured entities (species/state/district/person/etc).
- species_stats, source_stats, district_stats: trend and quality aggregates.
- sync_logs: sync cycle performance tracking.
- audit_logs: security/ops action trail.
- external_signals and watchlists: OSINT signal integration.

9. Live High-Risk Alerts section (deep explanation)

The Live High-Risk Alerts card is not a list of all incidents. It is a focused queue of incidents that crossed alert conditions.

9.1 How an alert is created

- Collector checks trigger reasons after intelligence computation: high risk threshold, priority species, organized network signal, repeat/spike signal.
- Severity is set to critical for strongest risk patterns; otherwise high.
- One alert per incident is enforced (deduplicated by news_id).

9.2 What the dashboard card shows

- Backend endpoint /api/alerts returns latest alert rows plus linked incident details.
- Frontend card title: Live High-Risk Alerts.
- It currently fetches recent alerts and displays up to 25 rows in the card for quick scanning.
- Each row includes title, risk pill, state/district/species, and source link.

9.3 Why this may not match total incidents

- Only incidents meeting alert triggers appear in this panel.

- Incident list includes many non-alert incidents too.
- Display and fetch limits intentionally cap rows for readability and performance.

10. Dashboard modules and what each one means

- KPI cards: Total Incidents, High Risk, States Active, Species Tracked, Sources Active, Source Reports Today.
- Map: geospatial spread of incident markers and hotspot intensity.
- Alerts: operational high-risk list for immediate attention.
- Analytics: timeline, top states, species distribution, source rankings (now deduplicated by unique incident).
- Incident Intelligence table: detailed per-incident analyst view.
- OSINT feed + recommendations: external signal context and common recommended actions.

11. Exports and officer reporting

- CSV export for spreadsheet analysis and data sharing.
- PDF export for briefing notes and official communication.
- Excel export for executive review.
- Excel 2-Sheet export: Total Incidents + Reports Today in one file.
- Briefing pack endpoint for structured machine-readable payloads.

Export routes now default to `min_confidence=0` so totals align with dashboard incident views unless a stricter filter is explicitly passed.

12. Security, access control, and reliability

- Frontend login for authorized users only; bearer token workflow.
- Admin logout endpoint for immediate session invalidation.
- Protected API routes under `/api` with admin checks.
- Rate limits for API, login attempts, and sync trigger endpoints.
- Audit log writing with retry/throttle behavior to handle DB lock pressure.
- Open-link endpoint `/open/{id}` resolves article links without write-path failures.

13. Deployment model (Railway + Vercel)

- Railway hosts backend service and scheduler (24/7).
- Vercel hosts React frontend.
- Volume at `/data` keeps DB, logs, and Excel persistent on Railway.
- CORS via `FRONTEND_ORIGIN` and API base URL via `VITE_API_BASE_URL`.

13.1 Operational recommendation

SQLite is practical for moderate traffic, but for higher concurrent writes and larger team load, PostgreSQL is recommended for stronger multi-user write performance.

14. End-to-end flow (one incident lifecycle)

- Article fetched from provider.
- Text normalized and analyzed by hybrid intelligence.
- Location/species/person details extracted.
- Dedup check runs against recent incidents.
- Incident inserted or merged; report_count/source_count updated.
- Alerts/reports/stats/entities updated.
- Frontend reflects update on next poll cycle.

15. Practical workflows for officers

15.1 Daily control-room workflow

- Open dashboard and monitor Live High-Risk Alerts first.
- Check map hotspots and top states for deployment focus.
- Use incident table filters by state/species/crime for unit-specific tasking.
- Export PDF/Excel for shift handover and superior briefing.

15.2 District investigation workflow

- Filter incidents by district/state and recent dates.
- Review involved persons, route hypothesis, and recommendations.
- Correlate repeated sources and report_count to identify sustained activity.
- Use alerts + source links as starting points for field verification.

16. API reference snapshot

Category	Key Endpoints
Dashboard	/api/dashboard-summary, /api/map-data, /api/chart-data, /api/filter-news
Incidents	/api/news, /api/live-incidents, /api/analyst-brief, /api/reports, /api/report/{id}
Alerts	/api/alerts, /api/alerts-popup
OSINT	/api/osint-feed, /api/external-signals, /api/trending-keywords, /api/watchlists
Exports	/api/export/csv, /api/export/pdf, /api/export/excel, /api/export/excel-incidents-reports
Security	/api/admin/login, /api/admin/logout, /api/security-status, /api/admin/audit-logs
Ops	/api/sync-status, /api/sync-history, /health, /open/{id}

17. Configuration quick guide

Variable	Meaning
TODAY_ONLY	true = strict today scope; false = use broader history/start-date scope.
START_FROM_DATE	Historical floor date in YYYY-MM-DD for scoped ingestion and views.
AI_THRESHOLD	Minimum confidence needed for baseline acceptance.
INDIA_THRESHOLD	Minimum India score for India relevance acceptance.
RISK_ALERT_THRESHOLD	Risk threshold used for high-risk alert behavior.
MAX_ARTICLES_PER_QUERY	Fetch depth per provider query.
MAX_QUERIES_PER_LANGUAGE	How many queries are rotated per language each cycle.
PROVIDER_PARALLEL_WORKERS	Parallelism level for provider fetch operations.
PROVIDER_MIN_REQUEST_INTERVAL_SECONDS	Swing between provider requests to avoid bans.
PROVIDER_RATE_LIMIT_COOLDOWN_SECONDS	Site limit from provider.
SUPPORTED_LANGUAGES	Languages included in multi-lingual search coverage.
ENABLED_PROVIDERS	Provider adapter list to use in sync cycles.

18. Known limits and best-practice guidance

- No internet monitoring system can guarantee 100% web coverage; this design maximizes practical coverage with multi-provider strategy.
- Aggressive search improves recall but can increase noisy candidates; monitor thresholds and watchlists regularly.
- For very high concurrent write loads, move from SQLite to PostgreSQL.
- Keep START_FROM_DATE and TODAY_ONLY configuration intentional to avoid scope confusion.

19. Suggested future enhancements

- Named Entity Recognition model for stronger multilingual person/place extraction.
- Entity resolution graph for aliases and network linking across incidents.
- Feedback loop from analyst corrections to tune confidence/risk calibration automatically.
- Advanced geocoding to district polygons and route path overlays.
- Automated daily officer briefing PDF generation by region/team.

20. Final summary

This project is an operational intelligence platform for wildlife-crime monitoring. It combines continuous search, AI-guided filtering, deduplication, risk prioritization, and field-usable outputs. For officials and officers, it reduces information overload, improves response speed, and creates a structured workflow for prevention, enforcement, and reporting.